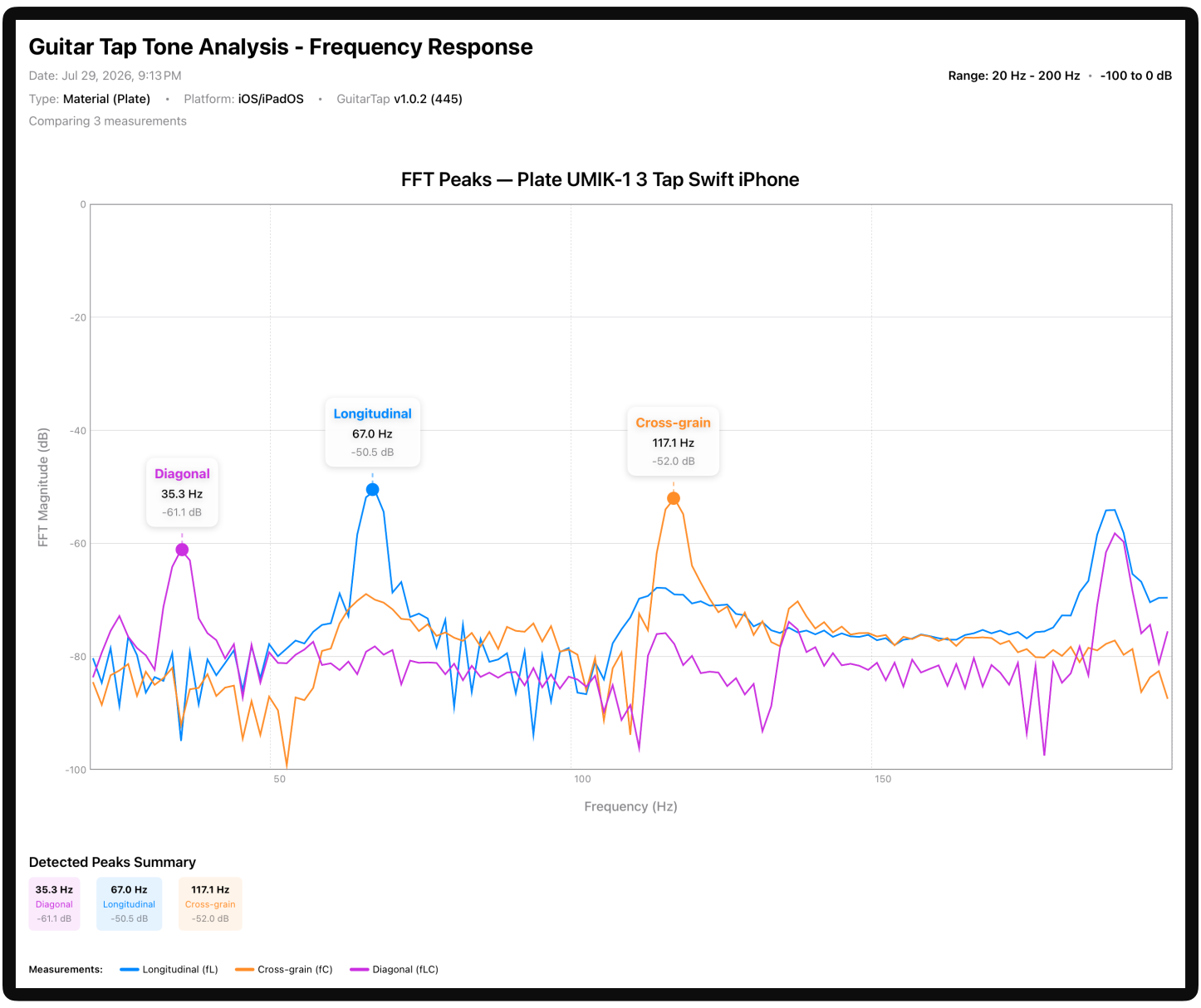
# GuitarTap

Tap Tone Analysis Report

Jul 29, 2026, 9:13 PM

| | |
|---|---|
| **Measurement Name:** | Plate UMIK-1 3 Tap Swift iPhone |
| **Type:** | Material (Plate) |
| **Frequency Range:** | 20.0 Hz – 200.0 Hz |
| **Microphone:** | Umik-1  Gain: 18dB · calibrated (7108913) |

## Frequency Spectrum




## Detected Peaks

| Frequency | Magnitude | Note | Q Factor | Role |
|---|---|---|---|---|
| 35.3 Hz | -61.1 dB | C#1 | 8.0 | Diagonal (fLC) |
| 67.0 Hz | -50.5 dB | C2 | 15.3 | Longitudinal (fL) |
| 117.1 Hz | -52.0 dB | A#2 | 20.0 | Cross-grain (fC) |

**Sample Dimensions**

Length: **557.50 mm** Width: **220.50 mm** Thickness: **4.85 mm**

Mass: **208.0 g** Density: **0.349 g/cm³**

**Body Dimensions**

Body Length (a): **490 mm** Lower Bout Width (b): **368 mm**

Panel Stiffness: **f_vs = 60 (Classical Top)**

**Gore Target Thickness**

**2.81 mm**

## Plate Properties

Speed of Sound (L): **4178 m/s**

Speed of Sound (C): **1142 m/s**

Young's Modulus (L): **6.09 GPa**

Young's Modulus (C): **0.45 GPa**

GLC (Shear Modulus): **0.340 GPa**

Cross/Long Ratio: **0.075**
*typical: 0.04–0.08*

Specific Modulus (L): **17.5** (Fair)

Specific Modulus (C): **1.3** (Very Good)

Radiation Ratio (L): **12.0**

Radiation Ratio (C): **3.3**

Long/Cross Ratio: **13.4**
*typical: 12–25*

**Overall Quality:** **Good**

**Three-Tap Measurement Process:**

- **1. Longitudinal (fL) Tap**
  Hold plate at 22% from one end along the length, near one long edge (not at the width node). Tap center.
- **2. Cross-grain (fC) Tap**
  Rotate 90°. Hold plate at 22% from one end along the width, near one short edge (not at the length node). Tap center.
- **3. Diagonal (fLC) Tap**
  Hold plate at the midpoint of one long edge. Tap near the opposite corner (~22% from both the end and the side). Measures shear stiffness.

*The strongest peak from each tap is auto-selected.*